

WASP-12b

R-2379

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_141221 · created 2026-08-20T01:12:18+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457012.6947 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.043
Transit depth (Rp/Rs)^2	2.39 +/- 1.33 [%]
Orbital Inclination (inc)	81.35 +/- 2.84
Airmass coefficient 1 (a1)	1.065 +/- 0.057
Airmass coefficient 2 (a2)	-0.05 +/- 0.029
Scatter in the residuals of the lightcurve fit is	5.44 %
Best Comparison Star	#2 - [252, 245]
Optimal Aperture	2.32
Optimal Annulus	7.1
Transit Duration (day)	0.1222 +/- 0.0087

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.043 vs 0.1178 (deviation 0.87σ)
Duration fit vs published	0.1222 d vs 0.125 d (deviation 0.32σ)
Mid-time O–C	-0.1 min
Depth SNR	3.6
Residual scatter	5.44 %

Light curve

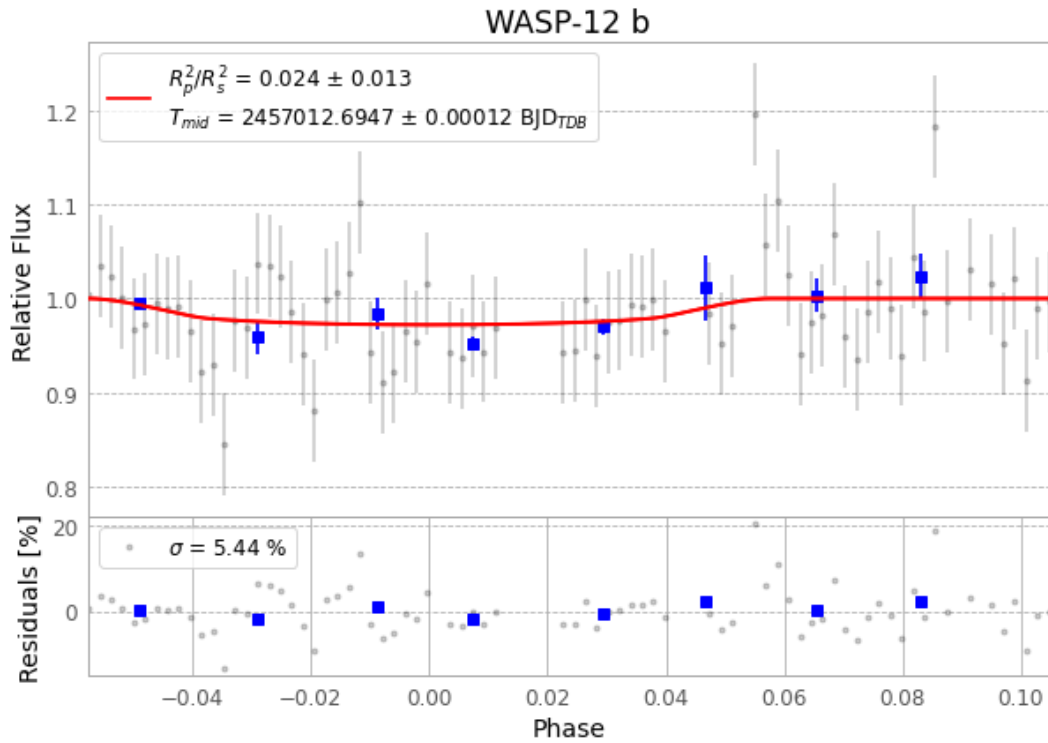


Figure 1 — FinalLightCurve_WASP-12 b_2014-12-20.png (SHA-256 0075c21bbd23a889...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [331, 208], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f027da4b1bcd5b3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

86 FITS frames (56 MB), WASP-12141221030012.FITS → WASP-12141221071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-141221.log (d0c5c203e0b9...), FinalLightCurve_WASP-12 b_2014-12-20.png (0075c21bbd23...), FinalLightCurve_WASP-12 b_2014-12-20.csv (c412cd59a26d...)

Compute resources

Wall time 138.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 01:12Z.